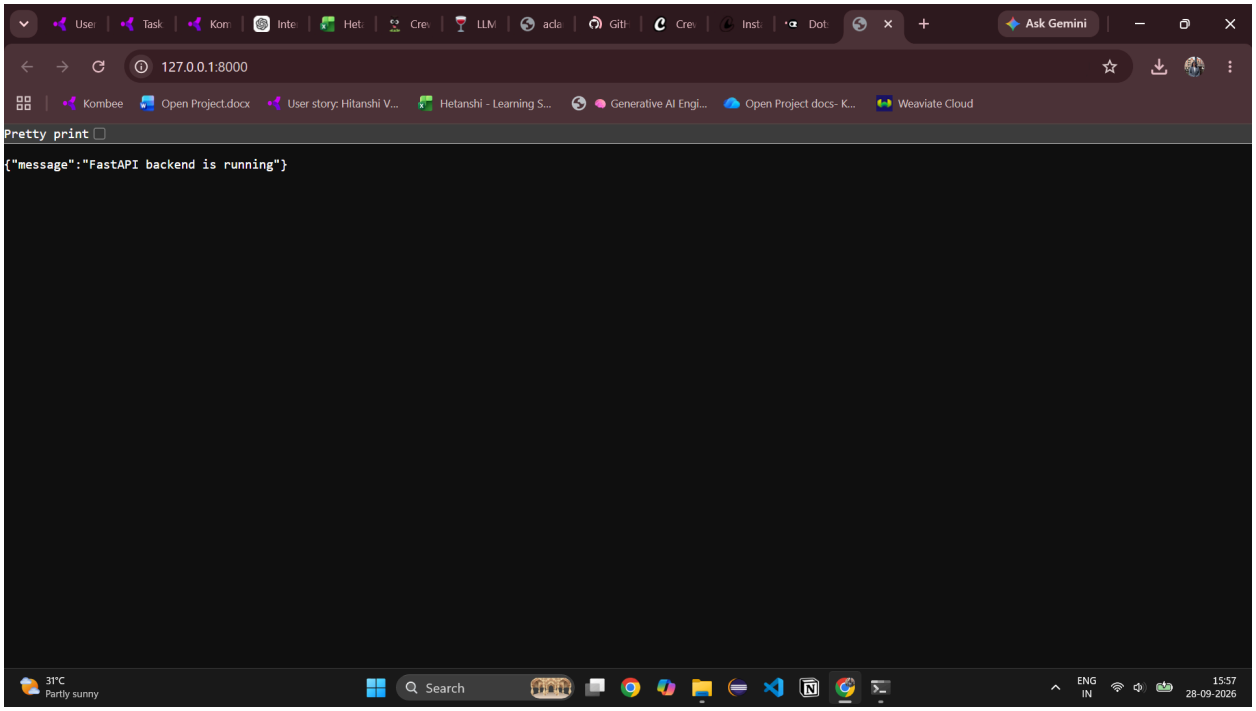
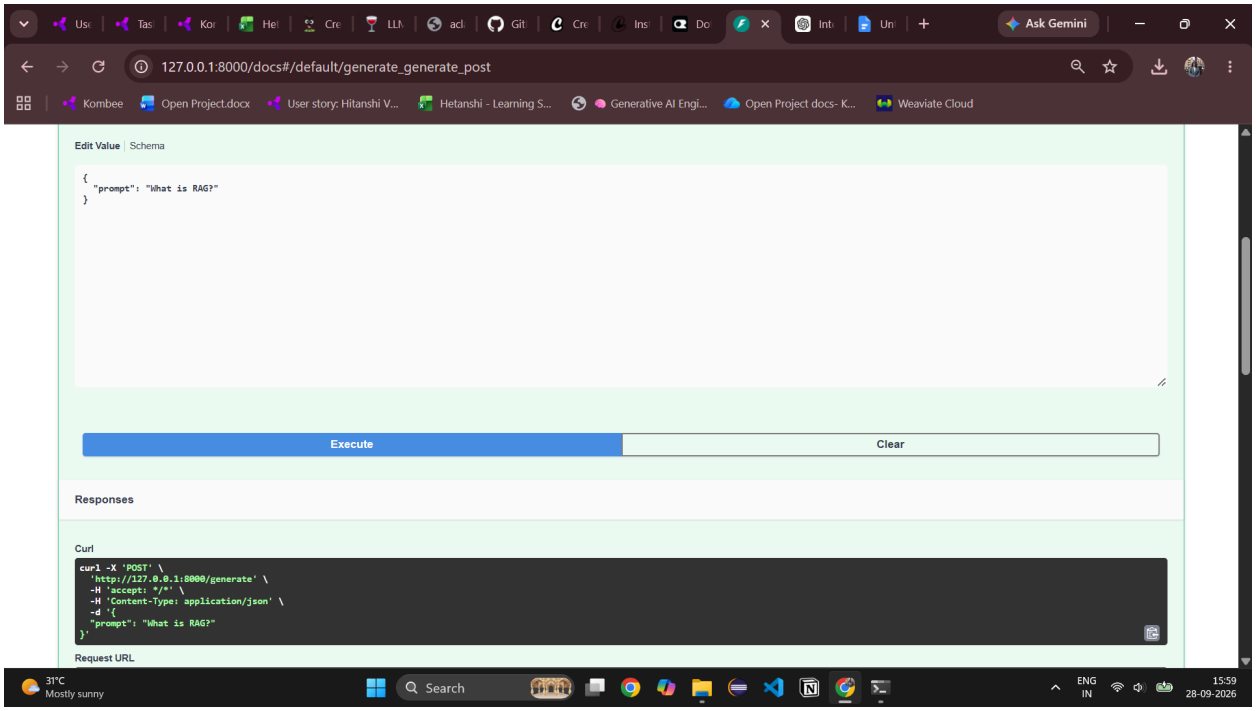
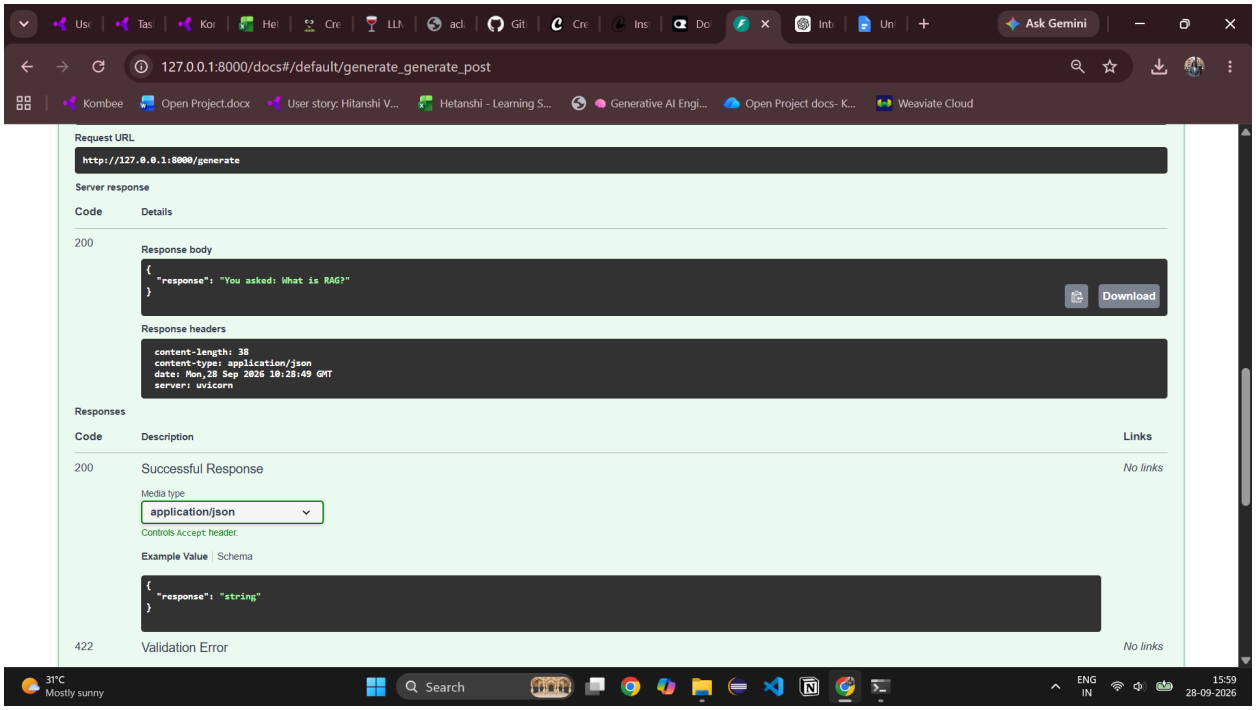
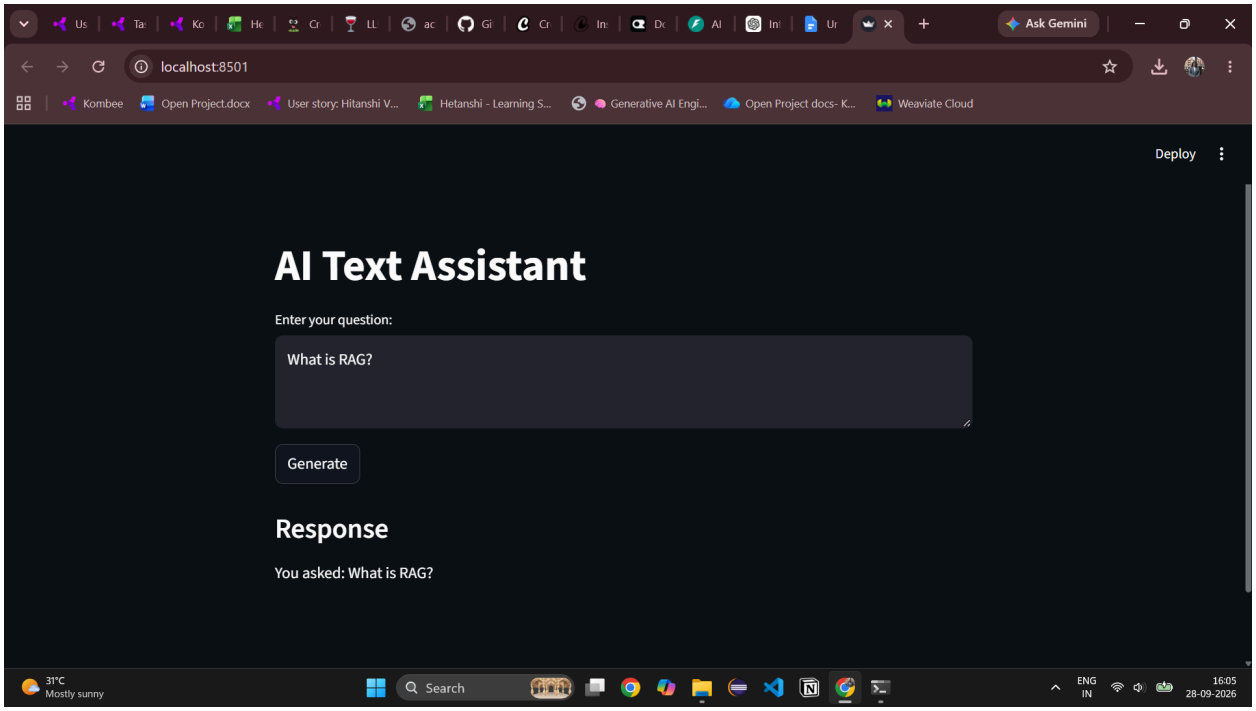
FAST API CONNECTED



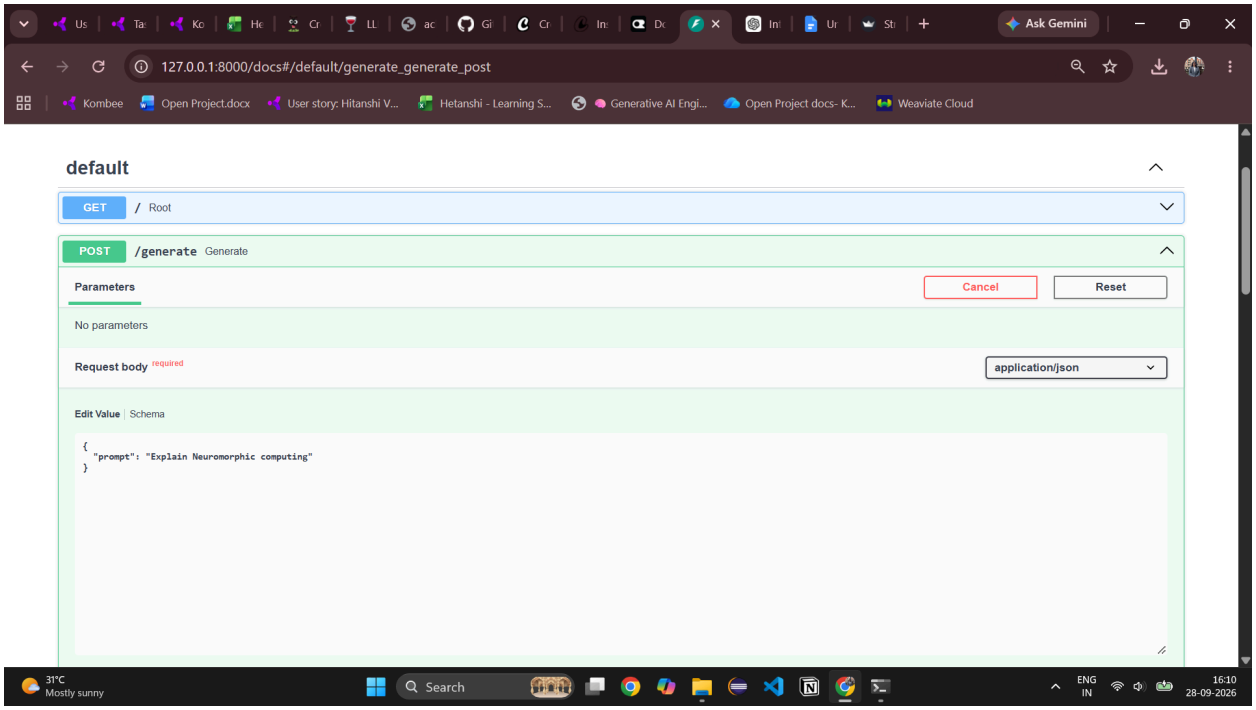


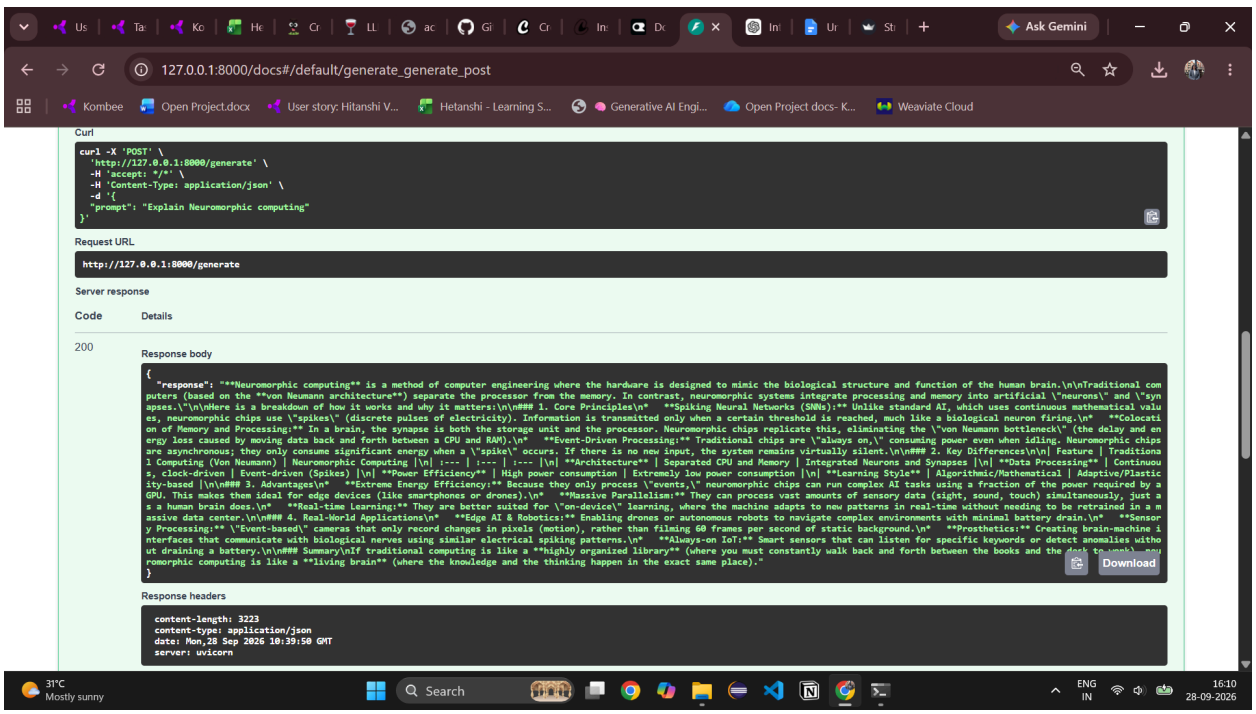


STREAMLIT UI RUNNING



TESTING FAST API WITH AI (Openrouter API)


127.0.0.1:8000/docs#/default/generate_generate_post
default
GET / Root
POST /generate Generate
Parameters
Cancel
Reset
No parameters
Request body required
application/json
Edit Value | Schema
{
"prompt": "Explain Neuromorphic computing"
}


127.0.0.1:8000/docs#/default/generate_generate_post
Curl
curl -X 'POST' \
'http://127.0.0.1:8000/generate' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
"prompt": "Explain Neuromorphic computing"
}'
Request URL
http://127.0.0.1:8000/generate
Server response
Code
Details
200
Response body
{
"response": "**Neuromorphic computing** is a method of computer engineering where the hardware is designed to mimic the biological structure and function of the human brain.\n\nTraditional computers (based on the **von Neumann architecture**) separate the processor from the memory. In contrast, neuromorphic systems integrate processing and memory into artificial \"neurons\" and \"synapses.\"\n\nHere is a breakdown of how it works and why it matters:\n\n### 1. Core Principles\n* **Spiking Neural Networks (SNNs):** Unlike standard AI, which uses continuous mathematical values, neuromorphic chips use \"spikes\" (discrete pulses of electricity). Information is transmitted only when a certain threshold is reached, much like a biological neuron firing.\n* **Colocation of Memory and Processing:** In a brain, the synapse is both the storage unit and the processor. Neuromorphic chips replicate this, eliminating the \"von Neumann bottleneck\" (the delay and energy loss caused by moving data back and forth between a CPU and RAM).\n* **Event-Driven Processing:** Traditional chips are \"always on,\" consuming power even when idling. Neuromorphic chips are asynchronous; they only consume significant energy when a \"spike\" occurs. If there is no new input, the system remains virtually silent.\n\n### 2. Key Differences\n\n| Feature | Traditional Computing (Von Neumann) | Neuromorphic Computing |\n| --- | --- | --- |\n| **Architecture** | Separated CPU and Memory | Integrated Neurons and Synapses |\n| **Data Processing** | Continuous, clock-driven | Event-driven (Spikes) |\n| **Power Efficiency** | High power consumption | Extremely low power consumption |\n| **Learning Style** | Algorithmic/Mathematical | Adaptive/Plasticity-based |\n\n### 3. Advantages\n* **Extreme Energy Efficiency:** Because they only process \"events,\" neuromorphic chips can run complex AI tasks using a fraction of the power required by a GPU. This makes them ideal for edge devices (like smartphones or drones).\n* **Massive Parallelism:** They can process vast amounts of sensory data (sight, sound, touch) simultaneously, just as a human brain does.\n* **Real-time learning:** They are better suited for \"on-device\" learning, where the machine adapts to new patterns in real-time without needing to be retrained in a massive data center.\n\n### 4. Real-World Applications\n* **Edge AI & Robotics:** Enabling drones or autonomous robots to navigate complex environments with minimal battery drain.\n* **Sensory Processing:** \"Event-based\" cameras that only record changes in pixels (motion), rather than filming 60 frames per second of static background.\n* **Prosthetics:** Creating brain-machine interfaces that communicate with biological nerves using similar electrical spiking patterns.\n* **Always-on IoT:** Smart sensors that can listen for specific keywords or detect anomalies without draining a battery.\n\n### Summary\nIf traditional computing is like a **highly organized library** (where you must constantly walk back and forth between the books and the desk to work), neuromorphic computing is like a **living brain** (where the knowledge and the thinking happen in the exact same place)."
}
Download
Response headers
content-length: 3223
content-type: application/json
date: Mon,28 Sep 2026 10:39:50 GMT
server: uvicorn

TESTING AI ON STREAMLIT UI

